

Week 9 Class Meeting — Summary

Learning Objectives

- Describe the importance of cost functions and their role in managerial decision-making.
- Demonstrate the relationship between a firm's production technology and its cost structure, and identify different types of cost.
- Explain how profit-maximizing firms behave in competitive markets, and how firm-level decisions aggregate into market supply.

Cost Functions & Production

- Cost functions are derived from the firm's production function. Short-run cost splits into fixed and variable components; marginal cost (MC) is the added cost of producing one more unit.
- Average cost (AC) and average variable cost (AVC) curves are U-shaped, and the marginal cost curve crosses both at their minimum points — the instructor walked through this standard relationship with a numeric example.
- The worked example referenced production-cost figures in roughly the \$10,000–\$78,000 range with about \$18,000 in additional cost, used to compute marginal/average cost from a production function. Audio quality was poor here (offline transcription only — see note below), so treat the specific dollar figures as low-confidence until checked against the Homework Module 9 problem set.

Profit Maximization in Competitive Markets

- In a perfectly competitive market, price settles at the minimum point of the (long-run) average cost curve — the zero-economic-profit benchmark.
- Firms maximize profit by producing where price equals marginal cost; if price falls below average variable cost, shutting down is better than operating at a loss in the short run.
- When price sits above minimum average cost, firms earn positive economic profit, which draws in entry (and the reverse triggers exit) until the market returns to zero-profit equilibrium — illustrated with a shortage/surplus example.

Profit Concepts (Setting Up the Discussion Post)

- Accounting profit: revenue minus explicit (accounting) costs only.
- Economic profit: revenue minus both explicit and implicit (opportunity) costs.
- Normal profit: the zero-economic-profit benchmark — the return just sufficient to keep resources in their current use.
- The instructor connected this to behavioral economics: real entrepreneurs' decisions reflect actual (bounded, psychological) mental processes rather than the purely rational profit-maximizing assumption — framed as this week's discussion-post question.

Assignments & Logistics

- Homework Module 9 covers production function, cost function, and profit-maximization problems (instructor referenced roughly 4 problems).
- Discussion post: initial post due Wednesday, with response posts due after, following the course's usual weekly cadence. Exact dates were not clearly audible — verify against the course site.